

Assignment: AWS Route53 Clone

Objective

Build a functional clone of the AWS Route53 web application with persistent storage and a backend API. The focus is on recreating the Route53 user experience and core workflows rather than implementing actual DNS functionality.

Note: Your application should match the original application in terms of UI and UX, look and feel should be exactly the same.

Tech Stack

Frontend: Next.js (TypeScript)

Backend: FastAPI

Database: SQLite

Scope

Authentication

Implement a simple mocked authentication system.

- Login
- Logout
- Session persistence

IAM, AWS Accounts, Organizations, Billing, and other AWS dependencies can be mocked.

Hosted Zones

Implement full CRUD functionality for Hosted Zones.

Users should be able to:

- View Hosted Zones

- Search Hosted Zones
- Create Hosted Zones
- Edit Hosted Zones
- Delete Hosted Zones

All data must persist in SQLite.

DNS Records

Implement full CRUD functionality for DNS Records within a Hosted Zone.

Support common Route53 record types such as:

- A
- AAAA
- CNAME
- TXT
- MX
- NS
- PTR
- SRV
- CAA

Users should be able to:

- View Records
- Search Records
- Create Records
- Edit Records
- Delete Records

All data must persist in SQLite.

Route53 Experience

The application should closely resemble the AWS Route53 experience, including:

- Navigation structure
- Hosted Zone management
- DNS Record management
- Tables
- Forms
- Search
- Filters
- Pagination

- Modals
- Notifications

The goal is to make the application feel like Route53 rather than a generic CRUD application.

Mocked Sections

The following sections can be present as placeholders:

- Dashboard
- Traffic Policies
- Health Checks
- Resolver
- Profiles

A simple "Coming Soon" page is sufficient.

Bonus (Optional)

- Import DNS records from BIND zone files
 - Export Hosted Zones as JSON or BIND format
 - Dark Mode
 - Keyboard Shortcuts
 - Bulk Operations
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Deliverables

Source Code

GitHub repository containing:

- frontend/
- backend/

Documentation

README containing:

- Setup instructions
- Architecture overview
- Database schema

- API overview

Demo: A hosted working link

Evaluation Criteria

- UI similarity to Route53
- Frontend engineering quality
- Backend/API design
- Database design
- Code quality and maintainability
- Documentation
- Overall completeness